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Impact of Diabetes on the Gut and Salivary IgA Microbiomes

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ABSTRACT Mucosal surfaces like those present in the lung, gut, and mouth interface with distinct external environments. These mucosal gateways are not only portals of entry for potential pathogens but also homes to microbial communities that impact host health. Secretory immunoglobulin A (SIgA) is the single most abundant acquired immune component secreted onto mucosal surfaces and, via the process of immune exclusion, shapes the architecture of these microbiomes. Not all microorganisms at mucosal surfaces are targeted by SIgA; therefore, a better understanding of the SIgA-coated fraction may identify the microbial constituents that stimulate host immune responses in the context of health and disease. Chronic diseases like type 2 diabetes are associated with altered microbial communities (dysbiosis) that in turn affect immune-mediated homeostasis. 16S rRNA gene sequencing of SIgA-coated/uncoated bacteria (IgA-Biome) was conducted on stool and saliva samples of normoglycemic participants and individuals with prediabetes or diabetes ($n = 8$/group). These analyses demonstrated shifts in relative abundance in the IgA-Biome profiles between normoglycemic, prediabetic, or diabetic samples distinct from that of the overall microbiome. Differences in IgA-Biome alpha diversity were apparent for both stool and saliva, while overarching bacterial community differences (beta diversity) were also observed in saliva. These data suggest that IgA-Biome analyses can be used to identify novel microbial signatures associated with diabetes and support the need for further studies exploring these communities. Ultimately, an understanding of the IgA-Biome may promote the development of novel strategies to restructure the microbiome as a means of preventing or treating diseases associated with dysbiosis at mucosal surfaces.

KEYWORDS IgA biome, microbiome, diabetes

Already a global epidemic, the prevalence of type 2 diabetes is expected to increase precipitously in the coming decades (1). Worldwide, 382 million people were living with diabetes in 2013, but projections predict 600 million by 2035 (2). Type 2 diabetes increases the risk of chronic inflammation, immune dysregulation, stroke, neuropathy, other diabetes complications, and certain cancers (3–11). Evidence suggests that the composition of the gut microbiome can influence inflammation and host metabolism in a manner consistent with what is seen at the onset of type 2 diabetes, intimating a possible causal role of the gut microbiome in type 2 diabetes progression (12, 13). An understanding of how the gut microbiome changes across time accompanying disease progression will provide insights into the etiology of type 2 diabetes and possibly identify novel diagnostic strategies and therapeutics. The challenge is how to identify

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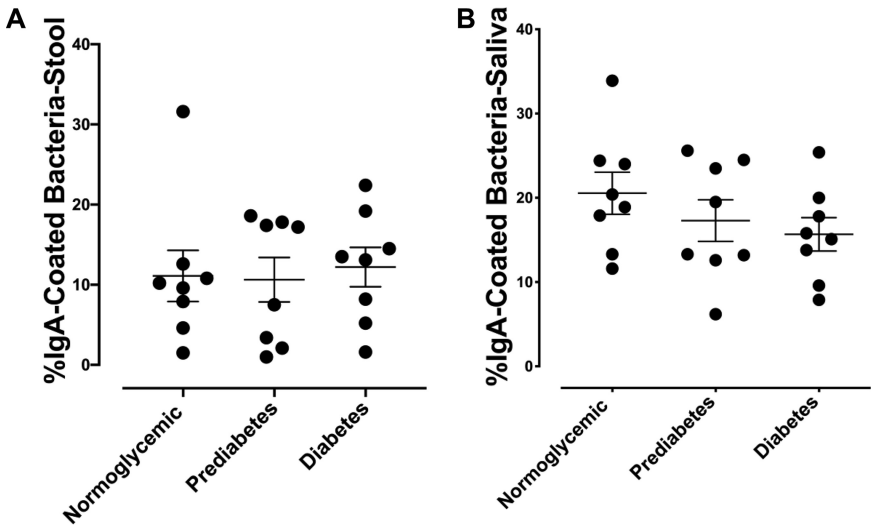


FIG 1 Percentages of IgA-coated bacteria in stool (A) and saliva (B) according to diabetes status. No significant differences in percent coating were observed within groups. Each dot represents a study participant. Horizontal bars represent the means ± standard errors.

among the hundred trillion (14) gut bacteria those species profiles most closely associated with health and disease. We addressed this challenge by analyzing changes in the immunoglobulin A (IgA) microbiome (IgA-Biome) of stool and saliva that accompany changes in glycemia as described previously.

Secretory immunoglobulin A (SIgA) is the single most abundant acquired immune component secreted onto mucosal surfaces and shapes the architecture of mucosal microbiomes (15, 16). Although present at all mucosal surfaces (e.g., in the gut, lungs, mouth, and eye), most SIgA is secreted into the gut (15), where it prevents direct interactions between the host intestinal mucosa and antigens derived from food, bacteria, or other sources (16). This process, known as immune exclusion (17–21), prevents antigen attachment through the agglutination of SIgA-bound targets that are subsequently removed by peristaltic and mucociliary movements (22–24). SIgA also mediates immune inclusion, a process that involves linking commensal microbiota constituents to the gut mucosa via glycans present on the SIgA secretory component (16).

The present study examines the IgA-Biome of the stool and saliva of normoglycemic, prediabetic, and diabetic individuals as a means of identifying microbial SIgA signatures associated with these diabetes phenotypes. Remarkably, while altered serum IgA responses have been observed in those with type 2 diabetes (7, 8, 25, 26), little is known about SIgA in the gut in the context of diabetes or hyperglycemia. An understanding of the SIgA-coated repertoire will identify microbial constituents that stimulate the host's immune response in the context of either health or disease. Using flow cytometry, we analyzed the IgA-Biome of stool and saliva using 16S rRNA gene sequencing. Characterization of the fecal and salivary IgA-Biome microbial signatures may yield diagnostic markers of diabetes and represent novel targets for the manipulation of the microbiome to reduce chronic inflammation and reverse or prevent diabetes progression.

RESULTS

Percentage of immunoglobulin-coated bacteria from stool and saliva. Bacteria present in stool and saliva samples collected from normoglycemic individuals ($n = 8$) or those with prediabetes ($n = 8$) or diabetes ($n = 8$) were incubated in the presence of fluorescently labeled anti-human IgA and/or anti-human IgM and sorted by flow cytometry. IgM coating was examined since this isotype can also be translocated to mucosal surfaces (14). In the stool samples examined, 11.1%, 10.63%, and 12.23% of bacteria from samples collected from normoglycemic, prediabetic, and diabetic individuals, respectively, were coated with IgA (Fig. 1A). The percentages of bacteria coated

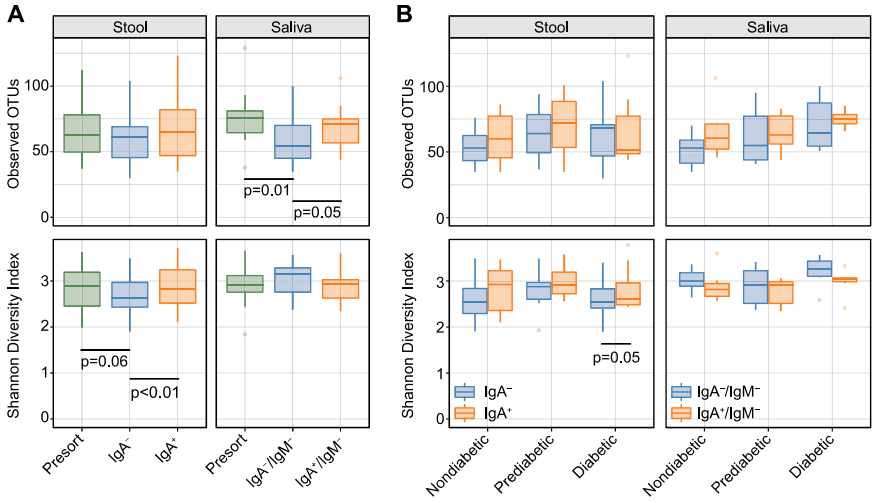


FIG 2 Alpha diversity of the stool and salivary IgA-Biomes. (A) Box plots depicting sample richness (observed OTU) and evenness (Shannon diversity index) among collective presort and sorted samples in both stool and saliva. (B) Same as panel A except that box plots indicate differences in diversity according to IgA-bound status stratified by diabetes phenotype. Only relationships with a *P* value of ≤0.05 by repeated-measures ANOVA are indicated.

with both IgA and IgM or IgM alone ranged between 0.1% and 1.35% (see Table S1 in the supplemental material). In the saliva, 20.55%, 17.3%, and 15.68% of bacteria from samples collected from normoglycemic, prediabetic, and diabetic individuals, respectively, were IgA positive (IgA^+)/IgM negative (IgM^-) (Fig. 1B). Although the flow cytometer successfully detected IgM^+ or IgA^+/IgM^+ bacteria, these organisms were not present in sufficient abundances to consistently allow for successful DNA extraction and 16S sequencing; therefore, these sort types were not included in subsequent analyses. Scatterplots for the respective stool and saliva samples used in subsequent analyses are shown in Fig. S1 to S3, and the percent coated IgM^+ or IgA^+/IgM^+ bacterial fractions are described in Tables S1 and S2. In addition, Tables S1 and S2 describe the number of bacteria used for 16S rRNA gene sequencing for each stool and saliva sort type analyzed.

Alpha and beta diversity. Changes in both alpha and beta diversity have been linked to changing glycemia profiles resulting in diabetes (27–29). Alpha diversity analysis, which evaluates within-sample diversity, did not identify differences in bacterial richness (observed OTUs [operational taxonomic units]) across stool IgA-Biome profiles ($P = 0.131$) (Fig. 2A). In comparison, differences in bacterial evenness (Shannon diversity index) were present ($P = 0.018$) and largely driven by decreased diversity in the IgA^- fraction (IgA^- versus IgA^+, $P = 0.003$; IgA^- versus presort, $P = 0.060$). Similar patterns were observed in stratified analyses (i.e., by glycemia status), although reductions in the evenness of the IgA^- versus IgA^+ stool microbiomes reached significance only among those with diabetes in this limited data set ($P = 0.049$) (Fig. 2B). These results suggest that while similar numbers (OTUs) of unique taxa are found in all stool IgA-Biome communities, the distribution (Shannon diversity index) of bacteria in the IgA^- fraction is more skewed to favor one or more types of bacteria; that is, the IgA^- fraction is less even than the presort or IgA^+ fraction.

In contrast to stool, saliva samples demonstrated discernible differences in bacterial richness ($P = 0.008$) (Fig. 2A), with the collective IgA^-/IgM^- fraction being less rich than the presort ($P = 0.013$) and IgA^+/IgM^- ($P = 0.050$) microbiomes. These observations extended to stratified analysis but were not statistically significant within any diabetes phenotype (Fig. 2B). No difference in bacterial evenness was evident across salivary IgA-Biome profiles ($P = 0.203$) (Fig. 2A), indicating that while the salivary IgA^-/IgM^- bacterial population is comprised of fewer unique OTUs than the IgA^+/IgM^- and presort populations, the communities are similarly distributed within individuals. The only sort type that

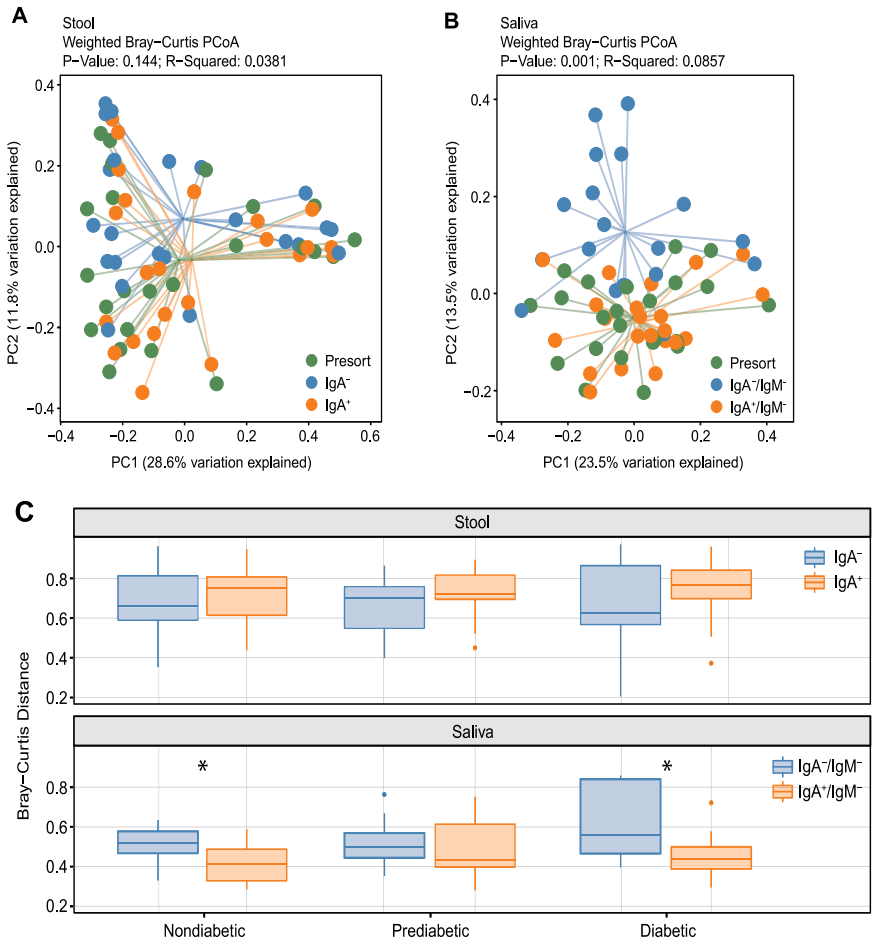


FIG 3 Beta diversity analyses of stool and salivary IgA-Biomes. (A and B) Differences in beta diversity by IgA-bound status were not evident in stool (A) but were observed in saliva (B). Beta diversity was measured by weighted Bray-Curtis distance and visualized by principal-coordinate analysis. Statistical significance was determined using PERMANOVA. (C) Box plots depicting weighted Bray-Curtis dissimilarities by IgA-Biome profile in stool (top) and saliva (bottom) according to diabetes phenotype. Significant differences were observed between IgA-coated and uncoated bacteria in saliva only. *, $P < 0.01$ by a Kruskal-Wallis test.

contained IgM-coated bacteria present at levels sufficient for 16S rRNA gene sequencing (Table S2) was IgA^+/IgM^+ bacteria sorted from saliva. However, alpha and beta diversity analyses as well as taxon abundance analysis did not identify significant differences between the IgA^+/IgM^- and IgA^+/IgM^+ populations (data not shown).

To investigate the beta (between-sample) diversity of the collective IgA-Biome communities, we visualized Bray-Curtis dissimilarity by principal-coordinate analysis (PCoA). PCoA plots revealed distinct clustering according to the IgA profile in saliva ($P = 0.001$) (Fig. 3B) but not stool ($P = 0.144$) (Fig. 3A). Notably, the salivary IgA^-/IgM^- microbiome was markedly different from both the IgA^+/IgM^- and presort microbiomes (IgA^-/IgM^- versus IgA^+/IgM^-, $P = 0.001$; IgA^-/IgM^- versus presort, $P = 0.001$), while the latter two communities exhibited considerable overlap ($P = 0.578$). Further analysis by diabetes status suggested that these observations may be driven by nondiabetic individuals (Fig. S4). We also examined intersample dissimilarity within each IgA-Biome signature by diabetes phenotype. IgA-coated communities in saliva were more similar to one another than uncoated populations, independent of diabetes status (Fig. 3C). These data collectively reinforce the importance of analyzing these distinct microbial communities in the context of microbiome-disease association studies.

IgA-Biome taxonomic composition. To identify bacterial taxa that discriminate the IgA-coated versus uncoated sorting fractions, we used the linear discriminant analysis

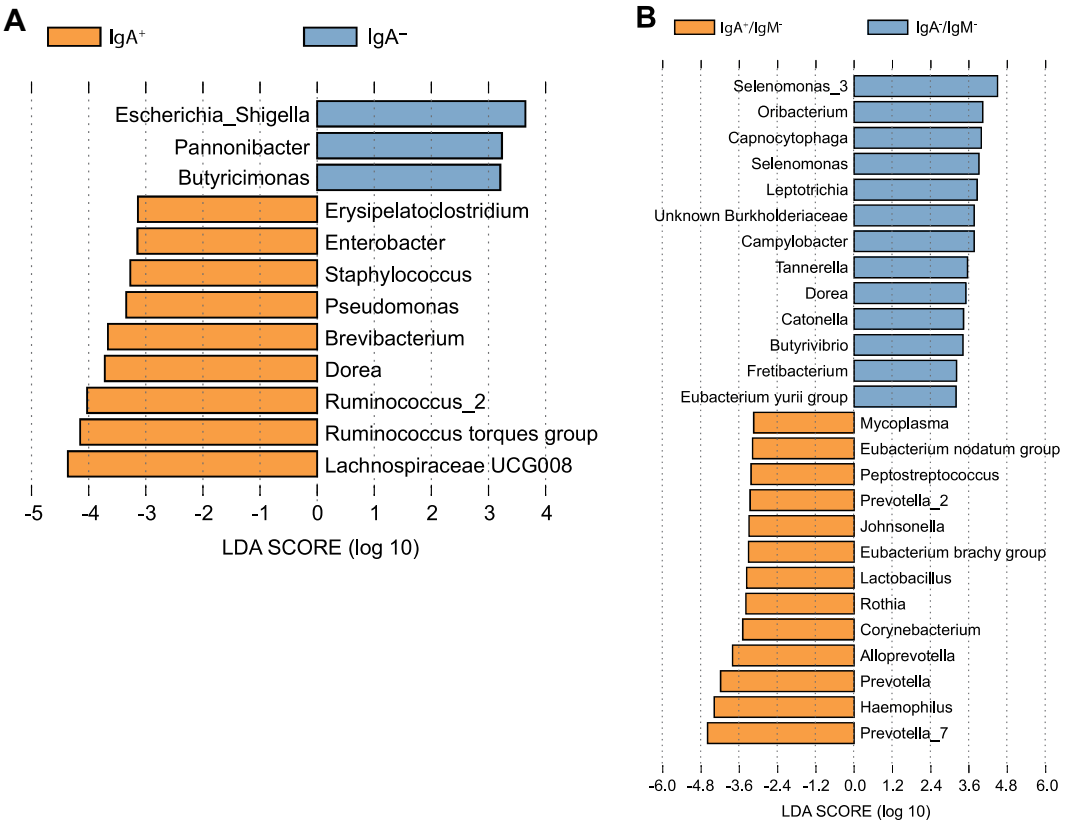


FIG 4 LEfSe identifies bacterial biomarkers associated with IgA-coated and uncoated genera in stool (A) and saliva (B). Analyses were conducted using parameters of an α value of <0.10 and a linear discriminant analysis (LDA) threshold value of ≥ 2.0.

(LDA) effect size (LEfSe) biomarker discovery tool (30). Among stool samples, *Ruminococcus_torques*_group, *Dorea*, *Pseudomonas*, and *Staphylococcus* were preferentially abundant in IgA$^+$ populations (all $P < 0.05$), while no genera characterized the IgA$^-$ fractions at the standard cutoff of an α value of 0.05. Using a less strict α value of 0.10, *Lachnospiraceae* UCG 008, *Ruminococcus*_2, *Brevibacterium*, *Enterobacter*, and *Erysipelatoclostridium* further distinguished IgA-bound communities, with *Escherichia_Shigella*, *Pannonibacter*, and *Butyricimonas* differentiating IgA$^-$ sorted bacteria (Fig. 4A).

In saliva, *Prevotella_7*, *Haemophilus*, *Prevotella*, *Corynebacterium*, *Peptostreptococcus*, and *Eubacterium_nodatum*_group characterized the IgA$^+$/IgM$^-$ fractions, while *Selenomonas*_3, *Oribacterium*, *Capnocytophaga*, *Selenomonas*, *Campylobacter*, *Catonella*, *Butyrivibrio*_2, and *Fretibacterium* differentiated IgA$^-$/IgM$^-$ communities (all $P < 0.05$) (Fig. 4B). Interestingly, *Dorea*, which characterized the IgA$^+$ fraction in stool, was found at a greater abundance in the IgA$^-$/IgM$^-$ population in saliva ($P = 0.050$). This observation was not explained by differences in OTU content as the same *Dorea* OTU dominated both sites, highlighting the potential for niche-specific IgA responses.

We next sought to determine whether discriminant taxa identified by LEfSe varied according to diabetes phenotype using the IgA coating index (ICI) (31). The ICI measures the enrichment of each genus in the IgA-coated versus the uncoated fraction, thereby accounting for intersubject variation in taxonomic abundances readily apparent in both stool and saliva (Table S3). Of those stool bacteria differentiating IgA-sorted fractions, none varied with diabetes status (Table S3). In contrast, the ICI scores of the discriminant salivary bacteria *Oribacterium* ($P = 0.035$), *Selenomonas* ($P = 0.007$), and, to a lesser extent, *Selenomonas*_3 ($P = 0.068$) increased with dysglycemia. Importantly, no significant differences in the abundances of these bacteria were evident among presort samples, which underscores the value of the IgA-Biome to inform previously unrecognized microbial relationships in the context of type 2 diabetes.

DISCUSSION

Recent efforts highlight the promise of the gut and oral microbiomes as targets in the development of novel prognostic, diagnostic, and therapeutic interventions. These microorganisms represent the bulk of the human microbiota (32), and as residents of mucosal microenvironments, they also face additional selection pressures imposed by the host's adaptive and innate immune systems (15). In the present study, we examined the stool and salivary microbiomes in the context of the host immune response by characterizing the IgA-Biome, which profiles bacteria based on their SIgA-bound status. Specifically, we investigated differences in the IgA-Biome in type 2 diabetes, a chronic inflammatory disease with known alterations of both the intestinal and oral microbiota (13, 32–36).

The mucosal immune system functions beyond simply protecting the host from infection. It also effectively serves as a "bouncer," selecting the residents of the intestinal microbiota as a means of maintaining gut homeostasis (16, 37, 38). To do this, it must decide between an active inflammatory response driven by either pathogen-derived (e.g., lipopolysaccharide and peptidoglycan) (39, 40) or self-derived (e.g., S100 proteins and heat shock proteins) (39–41) danger signals and a noninflammatory maintenance response governed by safety signals delivered by commensal bacteria and their by-products, healthy epithelial cells, and dietary components (37). Part of this decision-making process is established in early life, paradoxically through interactions between the developing immune system and the gut microbiota. This process is essential for both the development of $CD4^+$ and $CD8^+$ T cells as well as the maturation of B cells into IgA-secreting plasma cells (38).

Unlike traditional gut and oral microbiome sequencing approaches, the IgA-Biome offers a different perspective of the microbial communities colonizing mucosal surfaces. Specifically, it provides an understanding of host immune awareness of the myriad targets present in these ecological niches. Although bacteria found in stool are frequently used as a surrogate for the mucosal microbiome, data demonstrate that these two populations are distinct (42, 43). SIgA primarily targets microbes that come into contact with host tissues; therefore, the SIgA-coated fraction of the IgA-Biome is likely to be more similar to the mucosal microbiota than the microbiota colonizing the lumen of the intestine (44, 45). We hypothesized, therefore, that IgA-Biome profiles may be a better indicator of the mucosa-associated microbiota since these organisms will, by definition, have come into contact with the host's immune system and are thus more likely to have been in closer proximity to mucosal surfaces (16). The specificity of SIgA in the gut and at other mucosal sites (e.g., the oral mucosa, lungs, and eyes) dictates and shapes microbial community architecture, topography, invasiveness profiles, and the immunometabolome via antibody-mediated immune exclusion and inclusion (16, 44). The IgA-Biome signature therefore provides a snapshot of the organisms that have the attention of the host's immune system or, in the context of disease, organisms that should have the host's attention but for unknown reasons do not. Therefore, further characterizing the IgA-Biome at the species level will identify the microbial constituents that stimulate and interact with the host immune system to promote type 2 diabetes (31, 46–49).

In addition to the gut IgA-Biome, we examined IgA microbial signatures in saliva. The salivary microbiome represents a composite of various oral cavity niches (50) and has been linked to both oral and systemic diseases, including type 2 diabetes (32, 34, 51–53). Unlike the gut, the salivary microbiome appears more resistant to environmental stressors such as antibiotics (54). Zaura et al. hypothesized that this resiliency arose as a consequence of overcoming daily disruptions in the form of oral hygiene procedures as well as significant fluctuations in temperature and oxygen that do not occur in the gut (54).

Alpha diversity analyses of the stool IgA-Biome identified similarities in richness (OTUs); however, the presort and IgA^+ stool fractions were more even (Shannon diversity index) than the IgA^- fraction. In contrast, salivary IgA-Biome analyses revealed significant

differences in richness between the IgA^-/IgM^- compartment and the presort and IgA^+/IgM^- microbiomes. When parallel analyses were conducted comparing IgA-Biome profiles across diabetes phenotypes, statistical differences in richness were not evident, although a trend toward increased richness in the SIgA-coated compartment for both stool and saliva (independent of diabetes status) was observed. Shannon diversity analyses by diabetes status of the stool IgA-Biome showed a similar trend; that is, the SIgA-coated microbiota was more diverse (and statistically different in those with diabetes) than the uncoated fraction. Interestingly, the opposite trend was observed for the salivary IgA-Biome in that the SIgA-coated compartment was less diverse than the uncoated microbiota, although these differences were not statistically significant.

Beta diversity analyses of the IgA-Biome of stool and saliva identified significant differences in the salivary but not the stool microbiome using both PCoA and pairwise comparisons of Bray-Curtis dissimilarity values. Salivary IgA-Biome analysis demonstrated lower IgA^+/IgM^- intersample dissimilarity (i.e., Bray-Curtis dissimilarity closer to 0.0), indicating that the salivary IgA^+/IgM^- microbial communities, independent of diabetes status, were more similar to each other than those of the IgA^-/IgM^- microbiome, highlighting the importance of analyzing these distinct microbial communities in the context of mucosal microbiome studies.

Human studies of the gut microbiome and diabetes have largely focused on differences between individuals of varied glycemic statuses. *Bifidobacterium*, the most consistently identified taxon, showed an inverse relationship with hyperglycemia (55). Other bacteria negatively associated with prediabetes or diabetes are *Roseburia*, *Faecalibacterium*, and other *Clostridiales* members (56–59). Bacteria with frequent positive associations with type 2 diabetes include *Ruminococcus*, *Lactobacillus*, and various members of the *Proteobacteria* phylum (56–60). These relationships are not surprising given that *Proteobacteria* may result in a loss of gut barrier function, and hyperglycemia is itself associated with intestinal permeability (61). Of these taxa, we identified *Escherichia_Shigella* and *Pseudomonas* (*Proteobacteria*) to be associated with IgA^+ and IgA^- organisms, respectively, in addition to members of the order *Clostridia* (*Dorea* and *Ruminococcus*).

A similar analysis of the salivary IgA-Biome identified increased levels of *Rothia* and *Prevotella* in the SIgA-coated fraction. These genera were previously found to be associated with diabetes (62–65), while *Prevotella* has additionally been linked to poor oral health and increased body mass index (BMI) (50, 66). In contrast, *Alloprevotella*, also preferentially bound to SIgA, has been inversely associated with diabetes (53, 67). Taxa found to be differentially coated with SIgA included *Capnocytophaga*, *Leptotrichia*, and *Tannerella*, three genera previously linked to diabetes or impaired fasting glucose (62, 63, 67–71). *Selenomonas* and *Oribacterium* were also preferentially found among uncoated bacteria; however, the ICIs for these taxa varied by diabetes status. Specifically, prediabetic and diabetic individuals exhibited higher scores than nondiabetic individuals, indicating a greater propensity for these bacteria to be found in the IgA-coated fraction among those with dysglycemia. The relative abundance of these bacteria in presort samples did not vary with diabetes phenotype, demonstrating the potential for IgA coating profiles to identify taxa linked to changes in glycemia.

Identifying changes in the gut IgA-Biome that can be linked to transitions across glycemic profiles and to changes in IgA-Biome profiles at other mucosal sites, i.e., the oral IgA-Biome, may help identify subtle fluctuations associated with diabetes phenotypes and potentially identify IgA-Biome signatures that are shared between mucosal microenvironments. The presence of *Dorea*, for example, a genus associated with gut microbiota changes in those with diabetes, was identified in this study at both anatomical sites and in both the IgA-coated (stool) and -uncoated (saliva) compartments (57, 72). *Dorea* (and potentially other genera with similar profiles) may represent a candidate taxon that can be used to track changes in the SIgA response at different anatomical sites in the context of changing glycemia profiles.

The salivary IgA-Biome may provide a different avenue for the early detection of chronic disease states and serve as a potential alternative to the gut microbiome since

immune responses originating in the gut have implications at distant mucosal surfaces. For example, it was demonstrated that gut dysbiosis negatively impacted immune responses in the lung in response to *Mycobacterium tuberculosis* infection (73). Due to the low number of participants of each diabetes phenotype examined in this study, IgA-Biome data should be interpreted with caution. However, even with this limited sample size, differences between the IgA-Biome and the presort microbiome were observed. Future studies will employ whole-genome sequencing to identify organisms at the species level associated with changes in glycemia status in both the gut and saliva. These analyses will allow us to also test the hypothesis that changes in the gut IgA-Biome that accompany worsening glycemia are reflected by changes in the salivary IgA-Biome. These analyses would allow for the identification of salivary microbiome constituents that could serve as indicators of changes in the gut IgA-Biome.

Characterizing the IgA-Biome provides an awareness of taxa targeted by the host immune response and, equally important, those taxa not targeted in the context of specific disease states. Such studies can only further our understanding of disease etiology and may additionally promote the development of new modalities to restructure the microbiome as a means of preventing or treating diseases associated with dysbiosis at mucosal surfaces. Continued research of these unique signatures is greatly warranted.

MATERIALS AND METHODS

Study population and sample collection. Stool and saliva samples were collected from age-matched normoglycemic, prediabetic, and diabetic individuals (8 participants in each group) from Starr County, TX (Table 1). Subjects are participating in a larger study designed to identify changes in the nasal and gut microbiomes during the progression from normoglycemia to prediabetes to diabetes. This study was approved by the Institutional Review Board of the University of Texas Health Science Center, and informed written consent was obtained from each participant. Following enrollment, saliva samples (approximately 2 to 3 ml) were collected at the field office and immediately frozen at −80°C. Approximately half of the participants were able to provide a stool sample at the time of enrollment. Stool was collected using a feces catcher (Abbexa, Houston, TX), and a portion of the sample was placed into a Covidien precision stool collector and stored at −80°C. Participants unable to provide a stool sample at the field office collected samples in a similar fashion at home and contacted a Starr County Health Studies staff member when the sample was ready for pickup. Samples were stored at −20°C until they were transported to the field office, where they were stored at −80°C. Samples were shipped to the Houston laboratories on dry ice and stored at −80°C until processed for flow cytometry.

Diabetes classifications. Participants were defined as normoglycemic, prediabetic, or diabetic based on their fasting glucose levels, 2-h postload glucose levels, or percent glycated hemoglobin (HbA1c) (Table 1). A "normoglycemic" classification required all 3 measures to be within the normal range, while an abnormal value on any one test was sufficient to classify a subject as "diabetic."

Sample preparation for flow cytometry. Stool and saliva samples were prepared as described previously by Palm et al., with minor modifications (31). Stool samples were resuspended at a concentration of 0.1 g/ml in sterile-filtered, cold phosphate-buffered saline (PBS) (pH 7.4). Stool samples were allowed to rehydrate for 30 min on ice, and saliva samples were allowed to thaw at room temperature. One milliliter of either stool or saliva was then added to lysing matrix D tubes (MP Biologicals, Santa Ana, CA) and homogenized for 7 s using a BeadBeater (Biospec, Bartlesville, OK). Lysing matrix D tubes were then centrifuged at 730 rpm for 15 min, and 100 μl was collected for immunostaining. Each sample was washed three times with 1 ml cold PBS–1% bovine serum albumin (BSA; Sigma, St. Louis, MO). Between washes, samples were centrifuged at 9,230 rpm. After the first wash, 20 μl was removed and frozen at −80°C for use as the presort sample. After the last wash, samples were blocked in 25 μl of 20% mouse serum for 20 min on ice. Fluorophore-conjugated anti-IgA and anti-IgM antibodies were added to the respective samples at the concentrations specified by the manufacturers (listed below) for 30 min on ice. Samples were washed 3 times in PBS–1% BSA. After the last wash, samples were resuspended in 1 ml PBS–1% BSA and transferred to a flow tube. All centrifugation steps were carried out in a cold room at 4°C. Samples were maintained on ice until sorted.

Flow sorting of immunoglobulin-coated bacteria. Bacteria from stool and saliva samples were incubated with fluorescently labeled antibodies specific for either human IgA or IgM. Stool samples were sorted on two separate occasions, and saliva samples were sorted on a separate single day. Sorting was conducted using a BD SORP (special-order research product) FACS Aria II instrument (Becton, Dickinson, San Jose, CA), equipped with a forward-scatter photomultiplier tube (FSC-PMT) for improved small-particle detection, at the Baylor College of Medicine Cytometry and Cell Sorting Core Facility, where instruments go through daily quality assurance (QA)/quality control (QC) and NIH biosafety standards are followed. The instrument was set up for small-particle detection using both forward-scatter and side-scatter (SSC) thresholds to determine the signal from the background. Scatter voltage settings were determined using FSC height and SSC height parameters to adequately adjust signal-over-threshold values. The FSC-PMT in both area and height was used to verify FSC and SSC threshold settings. Typical singlet gating discrimination was not performed to prevent the exclusion of populations due to

TABLE 1 Participant characteristics by diabetes status[a]

Participant ID	Age (yr)	Sex	BMI (kg/m²)	Fasting plasma glucose level (mg/dl)		2-h postload glucose level (mg/dl)		HbA1c (%)
Normoglycemia								
Reference values				<100 mg/dl	and	<140 mg/dl	and	<5.7%
PM0004	49	Female	36.5	77		111		4.9
PM0028	53	Female	26.5	90		108		5.3
PM0030	52	Female	34.9	90		58		5.1
PM0033	38	Female	24.3	82		92		5.3
PM0052	52	Female	33.6	82		106		5.3
PM0054	51	Female	27	80		75		4.6
PM0062	38	Female	34.7	95		104		5.5
PM0073	44	Female	26.9	87		103		5.0
Mean (SD)	47.1 (6.3)		30.6 (4.8)	85.4 (6.1)		94.6 (18.8)		5.1 (0.3)
Prediabetes								
Reference value				100 ≤ mg/dl < 126	or	140 ≤ mg/dl < 200	or	5.7 ≤ % < 6.5
PM0011	43	Female	35.3	95		108		5.7
PM0017	51	Female	33.1	96		104		6.0
PM0022	38	Female	31.4	95		152		4.8
PM0027	55	Male	28.7	111		166		5.9
PM0040	53	Female	31	99		67		5.8
PM0041	52	Female	36.6	112		154		5.1
PM0044	45	Male	36.8	97		83		5.7
PM0056	50	Male	31	100		74		6.0
Mean (SD)	48.4 (5.8)		33.0 (3.0)	100.6 (7.0)		113.5 (39.0)		5.6 (0.4)
Type 2 diabetes								
Reference value				mg/dl ≥ 126	or	mg/dl ≥ 200	or	% ≥ 6.5
PM0001	54	Female	24.4	128		305		7.3
PM0013	51	Female	43.8	101		194		7.6
PM0019	41	Female	46.4	131		240		6.5
PM0043	53	Female	42.5	110		224		6.2
PM0079	41	Male	36.6	307		—[b]		12.3
PM0082	53	Female	42.5	123		147		6.6
PM0089	51	Female	29.9	122		218		6.5
PM0106	39	Female	35.2	127		273		6.7
Mean (SD)	47.9 (6.4)		37.7 (7.6)	143.6 (66.8)		228.7 (51.5)		7.5 (2.0)

[a]Shaded values indicate a measurement that qualifies an individual as prediabetic or diabetic based on the indicated cutoff values. Some individuals have more than one measurement that qualifies them as prediabetic or diabetic. Any measure of fasting glucose, 2-h postload glucose, or percent HbA1c can be used to classify individuals as having prediabetes or diabetes.

[b]A 2-h postload glucose test was not conducted on individuals with a fasting plasma glucose level of ≥160 mg/dl.

variations in scatters of bacteria. To reduce coincident events and sort error issues, the sample differential was maintained at the lowest setting, and the sample threshold rate was decreased by diluting the sample as needed with phosphate-buffered saline containing 1% bovine serum albumin. Background fluorescence was established using unlabeled specimens on all samples and examined in bivariate plots for the two channels of interest, fluorescein isothiocyanate (FITC) and phycoerythrin (PE), and then PE and allophycocyanin (APC), setting gates on bias to account for a range of autofluorescence. Interchannel compensation was done using single labeled samples of either conjugated antibody. Sorted samples were collected into chilled, prelabeled microcentrifuge tubes.

The first set of stool samples ($n = 4$ per glycemia group) was incubated with either mouse FITC-conjugated anti-human IgM (catalog number SA1-19598; Thermo Fisher, Waltham, MA) or PE-conjugated mouse anti-human IgA (catalog number 130-093-128; Miltenyi Biotec Inc., Auburn, CA). Accordingly, bacteria were sorted as either IgA^+, IgA^-, IgM^+, or IgM^-. Only 2/4 IgM^+ sorts contained sufficient bacteria for the successful extraction of DNA at a concentration suitable for 16S rRNA gene sequencing. Therefore, these samples were not included in subsequent analyses. The second set of stool samples ($n = 4$ per group) was tested after the PE–anti-human IgA (catalog number 130-093-128) used in the first experiment was no longer available for purchase. Instead, a recombinant PE-conjugated anti-human IgA (catalog number 130-116-878; Miltenyi) was paired with an APC-conjugated anti-human IgM (catalog number 130-093-076; Miltenyi) in the same sample tube; accordingly, these stool bacteria were sorted as either IgA^+/IgM^-, IgA^+/IgM^+, IgA^-/IgM^+, or IgA^-/IgM^-. Comparisons of percentages of IgA-positive bacteria following incubation with the two different anti-IgA antibodies did not identify significant

differences in percent IgA coating; that is, the percentages of IgA^+ and IgA^+/IgM^- coated bacteria were similar (see Fig. S5 in the supplemental material). For analysis purposes, sequence data obtained from IgA^+ (first set of sorted stool samples) and IgA^+/IgM^- (second set of samples) sorts were combined into a single "IgA^+" group.

Saliva samples were analyzed using the PE-conjugated anti-IgA (catalog number 130-116-878; Miltenyi)/APC-conjugated anti-human IgM (catalog number 130-093-076; Miltenyi) antibody pair. In contrast to stool sample analyses, all saliva samples were incubated simultaneously with anti-IgA and anti-IgM, enabling dual-label identification.

Most stool and saliva IgA biome analyses were conducted on half a million sorted bacteria, although some sorts had fewer bacteria (Tables S1 and S2). IgM^+ bacteria in stool ($n = 12$) or IgA^+/IgM^+ or IgA^-/IgM^+ bacteria in stool ($n = 12$) and saliva ($n = 24$) were present in numbers insufficient to sort (Fig. S1 to S3 and Tables S1 and S2). Only IgA^+/IgM^+ bacteria sorted from saliva samples were suitable for 16S rRNA gene sequencing (Table S2).

Sorted samples along with sheath fluid controls were transported on ice and centrifuged at 4°C at 10,000 rpm for 15 min. The majority of the supernatant was aspirated, leaving approximately 100 μl, and then frozen at −80°C until it was processed for 16S sequencing (31).

16S RNA gene sequencing. Genomic DNA was extracted with the MagAttract PowerSoil DNA kit (Qiagen) according to methods adapted from ones used in the NIH Human Microbiome Project (74). The V4 region of the 16S rRNA gene was amplified by PCR using barcoded primers (515F and 806R) and sequenced on the MiSeq platform (Illumina) via the 2- by 250-bp paired-end protocol, targeting at least 10,000 reads per sample.

Demultiplexed read pairs were merged using USEARCH v7.0.1090 (75) with the following parameters: merged length of $\geq$252 bp, minimum 50-bp overlap with zero mismatches, and truncation quality of $>$5. Merged reads were further filtered against a maximum expected error rate of 0.05, and PhiX sequences were removed. Reads were clustered into operational taxonomic units (OTUs) at a similarity cutoff value of 97% using UPARSE (76) and underwent chimera removal via USEARCH (75). OTUs were mapped against an optimized version of the SILVA v132 database (77) containing only sequences from the V4 region of the 16S rRNA gene. To reduce the influence of potential contamination from the sorting process, OTUs identified in at least two of three sheath fluid samples were removed from the data set if their average relative abundance among presort samples was $<$50% of the sheath samples. This resulted in the removal of six OTUs representing *Acinetobacter*, *Bradyrhizobium*, *Methylobacterium*, *Pseudomonas*, *Tyzzerella*, and a *Microbacteriaceae* family member. Following the removal of putative contaminants, data were rarefied to 1,319 reads per sample (Fig. S6).

Statistical analysis. Agile Toolkit for Incisive Microbial Analyses 2 (ATIMA2), a stand-alone tool for microbial data exploration, was used as an integrated solution for analyzing and visualizing microbiome data (https://atima2.jplab.net/). ATIMA2 uses permutational multivariate analysis of variance (PERMANOVA) to evaluate differences in the overarching community composition (beta diversity), while comparisons of community dispersion were performed via the Mann-Whitney U test. Alpha diversity analyses were performed in STATA 16 using repeated-measures analysis of variance (ANOVA) and the paired *t* test as appropriate. Differentially abundant taxa by IgA-bound status were determined via LEfSe (78) using parameters of an α value of 0.10 and a minimum LDA value of 2.0. LEfSe was limited to taxa identified in at least 10% of samples. Finally, the ICI was calculated as relative abundance of IgA^+ bacteria/relative abundance of IgA^- bacteria (31). When taxa were detected in the IgA^+ fraction but not the IgA^- fraction, a pseudoabundance value of 0.0008 (the limit of detection) was applied to the denominator. Samples in which taxa were not detected in either fraction were excluded from the analysis. Differences in ICIs were tested using the Kruskal-Wallis test. Unless otherwise indicated, results are considered significant at a *P* value of $<$0.05.

Data availability. The 16S RNA sequencing data are available in NCBI under BioProject accession number PRJNA662100.

SUPPLEMENTAL MATERIAL

Supplemental material is available online only.

SUPPLEMENTAL FILE 1, PDF file, 2.2 MB.

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